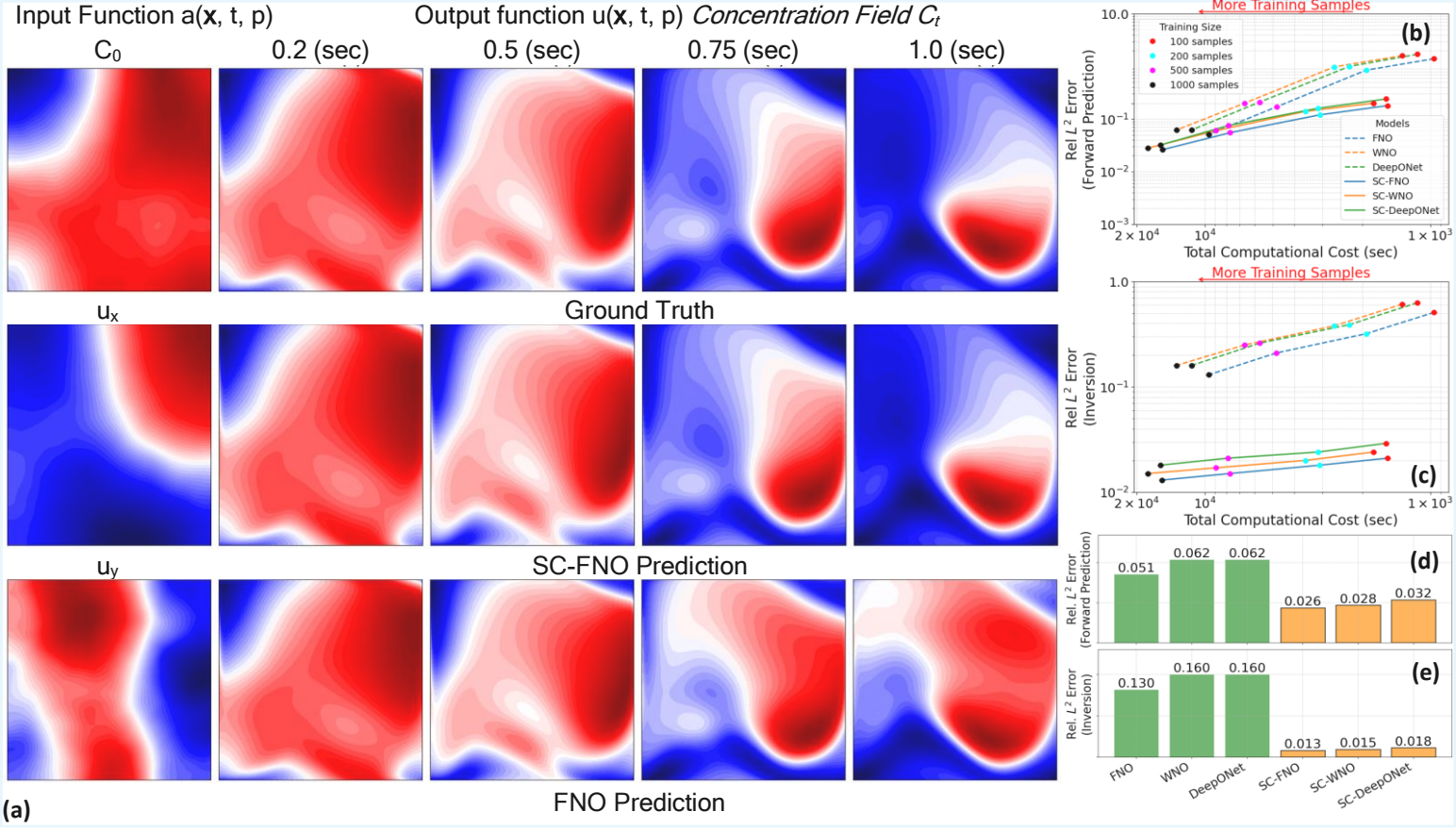
Input Function a(x, t, p)
Output function u(x, t, p) Concentration Field C_t
C_0
0.2 (sec)
0.5 (sec)
0.75 (sec)
1.0 (sec)
u_x
Ground Truth
u_y
SC-FNO Prediction
FNO Prediction
(a)
More Training Samples
(b)
Training Size
100 samples
200 samples
500 samples
1000 samples
Models
FNO
WNO
DeepONet
SC-FNO
SC-WNO
SC-DeepONet
Rel L^2 Error (Forward Prediction)
Total Computational Cost (sec)
More Training Samples
(c)
Rel L^2 Error (Inversion)
Total Computational Cost (sec)
(d)
Rel. L^2 Error (Forward Prediction)
0.051
0.062
0.062
0.026
0.028
0.032
(e)
Rel. L^2 Error (Inversion)
0.130
0.160
0.160
0.013
0.015
0.018
FNO
WNO
DeepONet
SC-FNO
SC-WNO
SC-DeepONet